Lesson 5.7 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Surface Area of Rectangular Prisms · I can find the surface area of a rectangular prism by adding the areas of its six faces.

S2: Surface Area of Triangular Prisms · I can find the surface area of a triangular prism using its net.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Surface Area of Rectangular Prisms

I can find the surface area of a rectangular prism by adding the areas of its six faces. · 6.G.A.4

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 6 × 9 = 54 ft² b. 2(12) + 2(8) + 2(6) = 52 cm²	Make an included-face checklist before summing areas.
2	2(8 × 5) + 2(8 × 2) + 2(5 × 2) = 80 + 32 + 20 = 132 in²	Make an included-face checklist before summing areas.
3	B — 2lw + 2lh + 2wh	Ask the learner to show why B — 2lw + 2lh + 2wh is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Includes a missing top or omits one of a matching pair..

Response: Make an included-face checklist before summing areas.

Also check: Multiplying l × w × h (volume) when surface area is asked.

Also check: Adding the three different faces without doubling them.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Find closed and open-top areas for a 4-by-3-by-2 m prism, top 4×3.

Teacher solution: Closed 52 m²; open 40 m².

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Closed: 2(15+10+6)=62 cm². Open: 62−15=47 cm². Count every included face exactly once.

Misconception response anchor: Includes a missing top or omits one of a matching pair.. Use make an included-face checklist before summing areas.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The relevant faces or regions are ___ because ___. My labeled dimensions show ___, so the measure is ___.

Session 2 · Surface Area of Triangular Prisms

I can find the surface area of a triangular prism using its net. · 6.G.A.4

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 2 × 6 = 12 m² b. 12 × 2 = 24 m²	Color or pattern-match each face on the solid to a term in the net sum.
2	Ends 2 × (½ × 10 × 12) = 120 in²; rectangles (10 + 13 + 13) × 5 = 180 in²; SA = 300 in²	Color or pattern-match each face on the solid to a term in the net sum.
3	D — the triangle's height	Ask the learner to show why D — the triangle's height is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Uses prism length as triangle height or slanted triangle side as height..

Response: Color or pattern-match each face on the solid to a term in the net sum.

Also check: Using the triangle's height as the side of a rectangle.

Also check: Counting only one triangle end.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A prism has 3 m–4 m–5 m right-triangle ends and length 10 m: find surface area.

Teacher solution: 12 + 120 = 132 m².

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Two ends: 2 × 1/2 × 3 × 4 = 12. Rectangles: (3+4+5) × 6 = 72. Total 84 cm². The 5 cm side is not the perpendicular height.

Misconception response anchor: Uses prism length as triangle height or slanted triangle side as height.. Use color or pattern-match each face on the solid to a term in the net sum.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The relevant faces or regions are ___ because ___. My labeled dimensions show ___, so the measure is ___.